

EARIN LAB 4 - Group 24, Variant 4

Regression and Classification

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Introduction

In this detailed report, we develop a predictive model to discern the survival outcomes of passengers from the Titanic disaster, utilizing a well-known dataset compiled for machine learning purposes. This task involves intricate data analysis to discern patterns and make predictions about survival based on passenger attributes. We employ two classification models—Logistic Regression and Random Forest Classifier—to establish which is more effective in this context, given the dataset's mixed data types and the binary nature of the response variable.

Data Preparation

The dataset encompasses several categorical and numerical variables, such as age, fare, sex, and class, each contributing uniquely to survival outcomes:

- **Missing Data Handling:** We imputed missing numerical data with median values to preserve distribution characteristics, while categorical missing data were filled using mode values to maintain the most frequent categories.
- **Categorical Variable Encoding:** We utilized one-hot encoding to transform categorical variables into a binary matrix, necessary for processing by machine learning algorithms.
- **Feature Scaling:** Numerical features were standardized using the StandardScaler to normalize their scales, preventing any single feature from disproportionately influencing the model due to its range.

This preprocessing ensured the dataset was primed for robust analysis and modeling, mitigating potential biases from raw data formats.

Model Definition

We defined the following models for this classification task:

1. **Logistic Regression:** Chosen for its baseline efficacy in binary classification tasks. It offers a probabilistic understanding of class memberships, which is essential for interpreting model outcomes in survival prediction.
2. **Random Forest Classifier:** Selected for its capacity to handle high-dimensional data and model complex dependencies without extensive preprocessing of the data. Its ensemble approach, which aggregates multiple decision trees to decide on the outcome, provides a higher level of accuracy and stability against overfitting.

Model Training and Evaluation

Using 4-fold cross-validation, we assessed the models based on accuracy, a suitable metric for binary classification tasks:

- **Accuracy as an Evaluation Metric:** Accuracy measures the proportion of total correct predictions (both true positives and true negatives) out of all predictions made. This metric was chosen because it directly reflects the model's ability to correctly identify survivors and non-survivors, aligning with our objective to maximize the correct identification of outcomes in a balanced dataset.

Cross-validation Results:

- **Logistic Regression:** [0.7697, 0.7753, 0.7865, 0.8371]
- **Random Forest:** [0.7584, 0.8258, 0.7584, 0.8371]

The Random Forest model demonstrated marginally higher consistency and performance across folds, leading to its selection for final testing.

Hyperparameter Tuning

We optimized the Random Forest model using a grid search, which systematically tested combinations of hyperparameters to find the configuration that maximized accuracy:

- **Tested Parameters:** Number of trees (**n_estimators**), tree depth (**max_depth**), minimum samples to split a node (**min_samples_split**), and minimum samples at a leaf node (**min_samples_leaf**).
- **Optimal Configuration:** Achieved with **n_estimators=100**, **max_depth=10**, **min_samples_split=2**, and **min_samples_leaf=1**.

This optimized model yielded a cross-validation accuracy of 0.82, reinforcing its suitability for this task based on our selected metric.

Final Model Testing and Analysis

Testing the optimized Random Forest on unseen data resulted in an accuracy of 86.03%, highlighting its effective generalization. The model not only performs well in recognizing patterns in the training data but also successfully generalizes these insights to new, unseen data.

Conclusion

This comprehensive analysis underscores the effectiveness of both Logistic Regression and Random Forest Classifier in predicting survival on the Titanic. The nuanced capabilities of the Random Forest, especially its handling of complex and non-linear relationships, render it slightly superior for this particular dataset. This project not only aids in understanding the survival factors during such disasters but also enhances our methodology in predictive modeling for binary outcomes.